

FRONT-END: Khởi tạo REACT JS

```
npm install
Windows PowerShell
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PS C:\Users\admin\Desktop\Chat-App> npm create vite@latest frontend

> npx
> create-vite frontend

|
|   Select a framework:
|   React
|
|   Select a variant:
|   JavaScript
|
|   Install with npm and start now?
|   Yes
|
|   Scaffolding project in C:\Users\admin\Desktop\Chat-App\frontend...
|
|   Installing dependencies with npm...
|
```

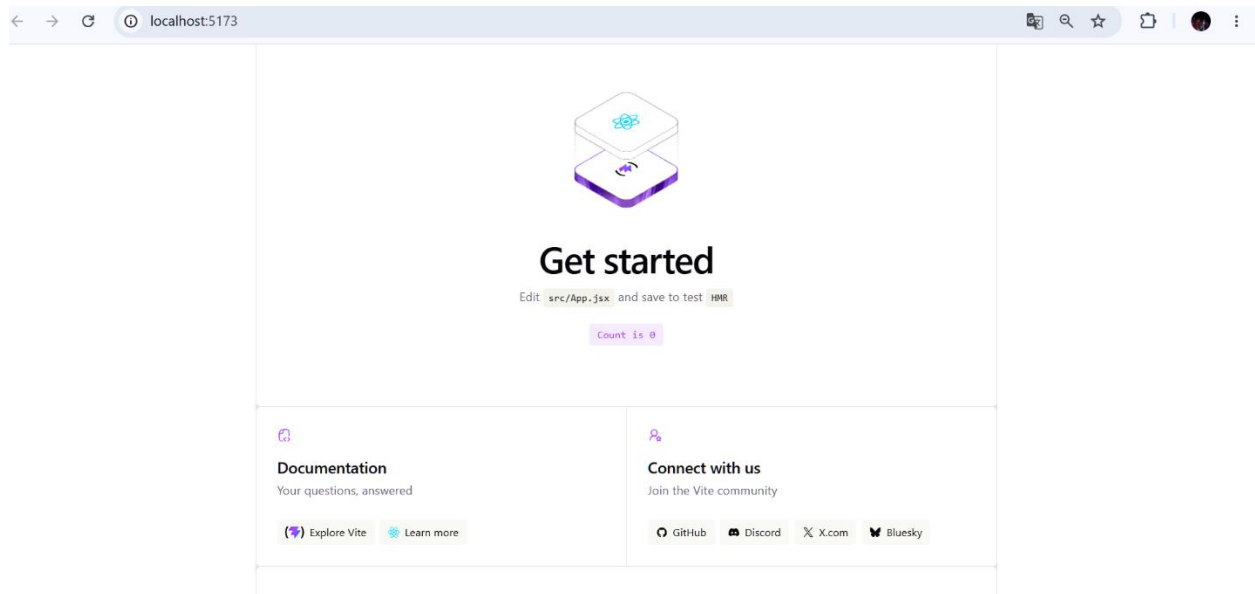
```
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|   Scaffolding project in C:\Users\admin\Desktop\Chat-App\frontend...
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```



BACK-END: Khởi tạo Fast API

```
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PS C:\Users\admin\Desktop\Chat-App> mkdir backend

Directory: C:\Users\admin\Desktop\Chat-App

Mode                LastWriteTime         Length Name
----                -
d-----         6/12/2026   1:59 PM                backend

PS C:\Users\admin\Desktop\Chat-App> cd backend
PS C:\Users\admin\Desktop\Chat-App\backend> python -m venv venv

PS C:\Users\admin\Desktop\Chat-App> cd backend
PS C:\Users\admin\Desktop\Chat-App\backend> python -m venv venv
PS C:\Users\admin\Desktop\Chat-App\backend> |
```

```
PS C:\Users\admin\Desktop\Chat-App> mkdir backend
```

```
Directory: C:\Users\admin\Desktop\Chat-App
```

Mode	LastWriteTime	Length	Name
d-----	6/12/2026 1:59 PM		backend

```
PS C:\Users\admin\Desktop\Chat-App> cd backend
```

```
PS C:\Users\admin\Desktop\Chat-App\backend> python -m venv venv
```

```
PS C:\Users\admin\Desktop\Chat-App\backend> .\venv\Scripts\Activate.ps1
```

```
(venv) PS C:\Users\admin\Desktop\Chat-App\backend> pip install fastapi uvicorn
```

```
Collecting fastapi
```

```
(venv) PS C:\Users\admin\Desktop\Chat-App\backend> pip install fastapi uvicorn
```

```
Collecting fastapi
```

```
  Downloading fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
```

```
Collecting uvicorn
```

```
  Downloading uvicorn-0.49.0-py3-none-any.whl.metadata (6.7 kB)
```

```
Collecting starlette>=0.46.0 (from fastapi)
```

```
  Downloading starlette-1.3.0-py3-none-any.whl.metadata (6.4 kB)
```

```
Collecting pydantic>=2.9.0 (from fastapi)
```

```
  Downloading pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
```

```
Collecting typing-extensions>=4.8.0 (from fastapi)
```

```
  Downloading typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
```

```
Collecting typing-inspection>=0.4.2 (from fastapi)
```

```
  Downloading typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
```

```
Collecting annotated-doc>=0.0.2 (from fastapi)
```

```
  Downloading annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
```

```
Collecting click>=7.0 (from uvicorn)
```

```
  Downloading click-8.4.1-py3-none-any.whl.metadata (2.6 kB)
```

```
Collecting h11>=0.8 (from uvicorn)
```

```
  Downloading h11-0.16.0-py3-none-any.whl.metadata (8.3 kB)
```

```
Collecting colorama (from click>=7.0->uvicorn)
```

```
  Downloading colorama-0.4.6-py2.py3-none-any.whl.metadata (17 kB)
```

```
Collecting annotated-types>=0.6.0 (from pydantic>=2.9.0->fastapi)
```

```
  Downloading annotated_types-0.7.0-py3-none-any.whl.metadata (15 kB)
```

```
  Downloading pydantic_core-2.40.4-cp314-cp314-win_amd64.whl (2.1 MB)
```

```
2.1/2.1 MB 104.9 kB/s 0:00:18
```

```
  Downloading annotated_types-0.7.0-py3-none-any.whl (13 kB)
```

```
  Downloading starlette-1.3.0-py3-none-any.whl (73 kB)
```

```
  Downloading anyio-4.13.0-py3-none-any.whl (114 kB)
```

```
  Downloading idna-3.18-py3-none-any.whl (65 kB)
```

```
  Downloading typing_extensions-4.15.0-py3-none-any.whl (44 kB)
```

```
  Downloading typing_inspection-0.4.2-py3-none-any.whl (14 kB)
```

```
  Downloading colorama-0.4.6-py2.py3-none-any.whl (25 kB)
```

```
Installing collected packages: typing-extensions, idna, h11, colorama, annotated-types, annotated-doc, typing-inspection
```

```
, pydantic-core, click, anyio, uvicorn, starlette, pydantic, fastapi
```

```
Successfully installed annotated-doc-0.0.4 annotated-types-0.7.0 anyio-4.13.0 click-8.4.1 colorama-0.4.6 fastapi-0.136.3
```

```
h11-0.16.0 idna-3.18 pydantic-2.13.4 pydantic-core-2.40.4 starlette-1.3.0 typing-extensions-4.15.0 typing-inspection-0.
```

```
4.2 uvicorn-0.49.0
```

```
[notice] A new release of pip is available: 25.3 -> 26.1.2
```

```
[notice] To update, run: python.exe -m pip install --upgrade pip
```

```
(venv) PS C:\Users\admin\Desktop\Chat-App\backend>
```

```
(venv) D:\Chat-App\backend>pip install "fastapi[standard]"
Defaulting to user installation because normal site-packages is not writeable
Collecting fastapi[standard]
  Using cached fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=0.46.0 (from fastapi[standard])
  Using cached starlette-1.3.0-py3-none-any.whl.metadata (6.4 kB)
Collecting pydantic>=2.9.0 (from fastapi[standard])
  Using cached pydantic-2.13.4-py3-none-any.whl.metadata (109 kB)
Collecting typing-extensions>=4.8.0 (from fastapi[standard])
  Using cached typing_extensions-4.15.0-py3-none-any.whl.metadata (3.3 kB)
Collecting typing-inspection>=0.4.2 (from fastapi[standard])
  Using cached typing_inspection-0.4.2-py3-none-any.whl.metadata (2.6 kB)
Collecting annotated-doc>=0.0.2 (from fastapi[standard])
  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting fastapi-cli>=0.0.8 (from fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard])
  Downloading fastapi_cli-0.0.24-py3-none-any.whl.metadata (6.4 kB)
```

```
(venv) D:\Chat-App\backend>pip install "fastapi[standard]"
Defaulting to user installation because normal site-packages is not writeable
Collecting fastapi[standard]
  Using cached fastapi-0.136.3-py3-none-any.whl.metadata (27 kB)
Collecting starlette>=0.46.0 (from fastapi[standard])
  Using cached starlette-1.3.0-py3-none-any.whl.metadata (6.4 kB)
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  Using cached annotated_doc-0.0.4-py3-none-any.whl.metadata (6.6 kB)
Collecting fastapi-cli>=0.0.8 (from fastapi-cli[standard]>=0.0.8; extra == "standard"->fastapi[standard])
  Downloading fastapi_cli-0.0.24-py3-none-any.whl.metadata (6.4 kB)
```

```
(venv) D:\Chat-App\backend\venv> python -m fastapi dev
```

```
FastAPI Starting development server 🚀

Searching for package file structure from directories with __init__.py files
Importing from D:\Chat-App\backend\venv

module 📄 main.py

code Importing the FastAPI app object from the module with the following code:

app Using import string: main:app

server Server started at http://127.0.0.1:8000
server Documentation at http://127.0.0.1:8000/docs

tip Running in development mode, for production use: fastapi run

Logs:

INFO Will watch for changes in these directories: ['D:\\Chat-App\\backend\\venv']
INFO Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO Started reloader process [15364] using WatchFiles
INFO Started server process [15080]
INFO Waiting for application startup.
INFO Application startup complete.
```

